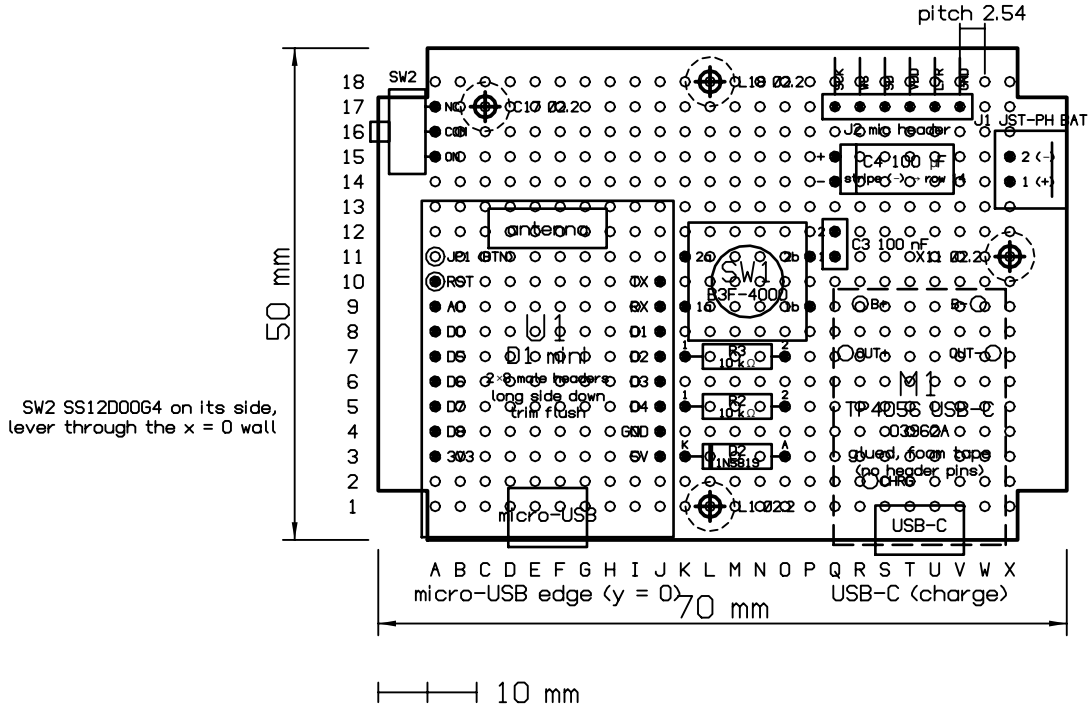
Perfboard 70 × 50 mm, COMPONENT side (top view), rev C

2.54 mm grid 24 × 18 holes Ø1.0 (A1 at 5.79, 3.41); 5 × 5 mm corner notches; mounting holes L1, L18, C17, X11 drilled Ø2.2 (Ø5 standoff dashed)



PARTS AND LEADS

U1	WEMOS D1 mini 3V3→A3 D8→A4 D7→A5 D6→A6 D5→A7 D0→A8 A0→A9 RST→A10 5V→J3 GND→J4 D4→J5 D3→J6 D2→J7 D1→J8 RX→J9 TX→J10
SW1	tactile 12×12, 7.3 mm actuator (B3F-4000) 1a→K9 1b→P9 2a→K11 2b→P11
D2	1N5819 Schottky, DO-41 A→O3 K→K3
R2	10 kΩ 1/4 W 1→K5 2→O5
R3	10 kΩ 1/4 W 1→K7 2→O7
C3	100 nF ceramic disc, 2.54 mm 1→Q11 2→Q12
C4	100 µF 10 V electrolytic Ø5×11 −→Q14 +→Q15
J2	1×6 right-angle male header SCK→Q17 WS→R17 SD→S17 VDD→T17 L/R→U17 GND→V17
J1	JST-PH 2-pin right-angle (S2B-PH-K-S) 1 (+)→X14 2 (−)→X15
SW2	SS12D00G4 slide switch (4 mm lever) ON→A15 COM→A16 NC→A17
M1	TP4056 USB-C charger module 03962A pads: B+, B−, OUT+, OUT−, CHRG (wired, see solder side)
JP1	solder bridge RST→A10 BTN→A11

mount Ø2.2 holes L1, L18, C17, X11; keep the 8 holes around each free (Ø5 standoff)

WIRING: see the solder side sheet (perfboard-solder-side) — no wires on this sheet.

Filled dots = leads soldered in grid holes; M1 is glued (foam tape) and wired from its 5 pads; JP1 = solder bridge A10–A11 (closed = battery mode).
BT1 LiPo lies on the enclosure floor under the board; M2 INMP441 sits under the lid, 6-wire lead to J2. Source: layout/layout.json rev C.